"Measuring Air Quality" Assignment Information and Instructions

The aim of this assignment is to collect meteorological data and examine it using statistical methods and error analyses. You will interpret and discuss the data collected during the practical.

The information below describes how to analyse data using MS Excel (using data that will be provided on Blackboard following the practical).

You will work individually to answer the questions and then answer the questions in the online test. The online test is not time-limited and you can repeat it as many times as you like, so that you can learn how to calculate the correct results.

You will be given instant automatic feedback on the online test but it is NOT assessed (does not count towards your degree mark).

Instructions:

Data: In the "Calibrated Data" folder within the air quality practical folder you will find a data file (an Excel Spreadsheet) containing calibrated kestrel data and data from the high-quality weather station placed in Brunswick Park). This may not be the same data as the data collected by your group. Please use this data for the analysis described below.

Make a note of your answers to the following questions, so that you can answer the same questions in the online test when you are ready. The online test will also include multiple choice questions about how to interpret the results you will calculate. You can practice your answers to those questions as many times as you like and feedback will be given when you submit your answers.

A YouTube video that describes how to calculate mean, standard deviation and standard error of measurement data using Excel is available in Assignment Information and Help Sheets folder. Please watch that video if you get stuck.

Question 1 - Calculate the mean (average) of temperature (in degrees Celsius) of the Kestrel data

Question 2 - Calculate the mean (average) of temperature (in degrees Celsius) of the Weather Station data

Question 3 – Calculate the offset (the difference) between your answers in questions 1 and 2 by subtracting your answer to question 2 from the answer to question 1. This is the offset. Can you think of what physical environmental and instrumental reasons might be responsible for this difference?

Question 4 - Calculate the standard deviation of temperature data (in degrees Celsius) of the Kestrel data. This gives an indication of the combined instrumental noise error (random error) and the natural variability in the atmosphere at the location where the data was collected.

Question 5 - Calculate the standard deviation of temperature data (in degrees Celsius) of the Weather station data.

Question 6 – Calculate the difference between your answers in questions 4 and 5 by subtracting your answer to question 5 from the answer to question 4. This is the difference in the variability measured by each instrument. Can you think of what and instrumental reasons might be responsible

for this difference (remembering that the weather station is expected to be more accurate than the Kestrel handheld instrument?

Question 7 – Calculate the standard error of the temperature data from the Kestrel data

Question 8 – Calculate the standard error of the temperature data from the weather station data

Question 9 – repeat the steps above (questions 1 to 8) for both pressure (in units of hPa) and relative humidity (%) and make a note of your answers.

Once you have finished, complete the online test in the "Assessment and Feedback – online test" folder in the air quality practical folder when you are ready. There is no time limit. You will receive instant feedback on your answers, along with the correct answer. You may retake the test again if you wish. Marks will not be used for assessment – this is a learning exercise that will help you to interpret data you may collect in the field in the future.

At the end of this assignment, you should be able to understand the sources of random error in any scientific measurement dataset, as well as the sources of variability in the real atmosphere, noting that it can sometimes be difficult to distinguish between them. In measurement science, the way to separate these two effects (one being a real change and the other being due to instrumental uncertainty) is to calibrate instruments in controlled conditions (e.g. in a laboratory where the temperature, pressure and humidity is kept constant). If they are held constant, then we can be sure that any variability we measure in those controlled conditions represents the instrumental random error alone. This would then allow us to separate the natural variability (in a real and changing atmosphere) from the instrumental uncertainty (random error) when we conduct measurements in the field.